

MEC4001 – Biomechanics

Assignment 1 – Gait Analysis

Analysing the lower-limb biomechanics during one gait cycle with respect to the joint angles , joint moments and muscle activation patterns.

Question 1

Provide a technical description of a gait cycle (including the different phases and sub-phases). From the data collected in the laboratory session, identify the best possible gait cycle and crop it. Insert gait cycle events and export the data to .CSV format. Give reasons why you chose that specific gait cycle and how you accurately chose the events. Report the frame numbers of the events. Plot the Pedotti diagram (z-component of the GRF) for one of the steps and discuss the shape of the diagram. Point out any deviations from the normal (if any).

Question 2

Describe the necessary apparatus used in the motion laboratory to collect the gait kinematics and kinetics, muscle motor unit action potentials and GRF during gait. Discuss the basic principle of these measuring instruments (i.e. explain how the biomedical instrumentation captures the data it is designed to collect).

Question 3

Describe the methodology undertaken during the experiment carried out. Mention any assumptions and precautions that were taken during the entirety of the experimental procedure.

Question 4

Provide gait graphs of the hip, knee and ankle angles and moments in the sagittal plane normalised over one gait cycle (12 in total). Identify and label the important stages of the gait cycle on the graphs in terms of % of the cycle when the event occurs. Discuss the shape of these graphs and relate the results to theoretical graphs found in literature (and the lecture notes).

Question 5

Derive the force profiles of the recorded muscles for one gait cycle. Explain how you derived the force generated by the muscles of interest for one gait cycle (include a clear explanation of the signal post-processing performed – give reasons why the techniques used in the signal post-processing were implemented for the recorded bio signals). Compare and discuss these graphs to theoretical profiles found in literature (and the lecture notes). [Note: Use the data collected during the MVC contraction of the Vastus Medialis to obtain the force-voltage relationship. Subsequently apply the relationship to all the sEMG profiles recorded during the gait cycle experiment].